

Component	Quantity	Notes / Optimization	Cost (INR)
Raspberry Pi Zero 2 W	1	Main controller	2,000
Raspberry Pi NoIR Camera v2	1	Compatible with Pi Zero	2,000
Micro SD Card (32 GB, Class 10)	1	OS + storage	600 – 800(opt)
Micro USB OTG Adapter + USB Hub	1	For keyboard/mouse during setup	300 – 500(opt)
20-Diopter Aspheric Lens	1	Optical magnification	2000
IR LED (850 nm / 940 nm) + 100 Ω resistor	2–3 LEDs + resistors	Standalone LEDs for retina illumination, simpler than sensor module	150 – 300
16×2 LCD Display (HD44780) + I2C backpack	1	Shows status messages, I2C simplifies wiring	400 – 500
Jumper wires (male–male & male–female)	1 set	Connect LEDs, LCD, and GPIO	200 – 300
Breadboard (half-size)	1	Solder-free prototyping	150 – 250
Power Adapter (5 V 2 A micro USB)	1	Stable power supply	300 – 500
Optional: Extra IR LEDs	2–3	Brighter/ uniform illumination	100 – 200

Approx 6500 - 7000 rs